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Grammatical understanding predicts reading comprehension in secondary-level students: insights from a Finnish national survey

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ABSTRACT

A lot of research into the relationship between explicit grammatical understanding and literacy development in L1 contexts has focused on writing. A lot less is known, however, about the interconnectedness of explicit grammatical understanding and reading. This study investigated the relationship between grammatical understanding and reading comprehension in Finnish 9th grade students. Using national assessment data ($N = 6{,}044$) and multilevel quantitative analyses, the study found that explicit grammatical understanding accounted for a significant portion of the variance in reading skills, even after controlling for various background factors such as school, family socioeconomic status, homework and book reading habits, attitudes towards studying L1, and writing ability. The findings suggest, without implying causality, that understanding language structure and making explicit metalinguistic observations are closely connected to reading comprehension. The study encourages further research on the benefits of explicit grammar teaching for reading comprehension using different study designs. Additionally, it highlights the need to investigate how grammar instruction can be linked to reading, as well as explore the effectiveness of such instruction in different educational and linguistic contexts.

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Grammatical understanding; metalinguistic understanding; reading comprehension; L1 education; secondary education

Introduction

Grammar teaching in L1 contexts has historically been thought of as an important part of language education for one of two reasons. Either grammar is seen as valuable in its own right, simply because language is an important phenomenon in our world that deserves to be understood (Hudson 2004; Van Rijt and Coppen 2021), or grammar is seen as a potential way of improving literacy development (Locke 2010; Fontich and Camps 2014). While the former perspective is increasingly recognized in the past years, a lot of questions remain about the relationship between L1 grammar education and students' literacy development: under what conditions might explicit grammar teaching

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contribute to gains in literacy, and how do teachers and learners deal with the relationships between grammar and literacy?

Most of the research examining relationships between L1 grammar and literacy has targeted students' writing. For decades, the general consensus seems to have been that explicit forms of grammar teaching (particularly decontextualized parsing) have a negligible or even harmful effect on writing skills (see e.g. Andrews et al. 2005; Graham and Perin 2007), although more recent studies suggest that contextualized grammar teaching can strongly improve writing ability (Fearn and Farnan 2007; Fontich 2016; Jones et al. 2013; Myhill et al. 2012; Myhill et al. 2018; Steenbakkers 2023), a finding that is reflected in a recent meta-analysis about writing (Graham et al. 2023). Some studies have also indicated that certain syntactic choices can influence the quality of written texts (Crossley 2020; Myhill 2008; Van Rijt et al. 2021, 2022) and that students' level of grammatical understanding predicts their writing skills (Marjokorpi 2023a).

The relationship between grammatical understanding and reading has been less researched within the field of L1 education (Bowey 1993; Van Rijt 2020, p. 13), even though there are important reasons to believe such a relationship exists. For example, psycholinguistic research has shown that reading requires syntactic processing (Verhoeven and Perfetti 2008; Verhoeven et al. 2011), and explicit training of such syntactic processing skills may therefore contribute to better reading. In this paper, we will examine these relationships based on a national assessment study, involving Finnish secondary school students ($N = 6{,}044$). Compared to, for example, Indo-European languages, the Finnish language is relatively agglutinative (Comrie 1989). This property raises the hypothesis that understanding grammar, and especially morphology, might play a significant role in understanding texts in Finnish language. If relationships exist between grammatical understanding and reading comprehension, this could have implications for the place of explicit grammar teaching in the language curriculum.

Defining the concepts: grammatical understanding and reading comprehension

Grammatical understanding

One of the difficulties in empirically examining the connections between grammatical understanding and reading comprehension has been the diverse way in which grammatical understanding has been operationalized. While research publications on morphological understanding have been clearer in defining their key concept, some studies have used the wide-ranging umbrella term 'grammatical understanding' or equivalents thereof to cover many kinds of metalinguistic knowledge and understanding. In this paper, we define grammatical understanding as 'one's ability to describe, comprehend and/or explain (aspects of) grammar, coherently using relevant grammatical (meta)concepts, terms and/or linguistic sources, grammatical repertoire, and argumentation' (Banga and van Rijt 2023). Grammatical understanding therefore requires a deep knowledge about the morpho-syntactic and phonological structures of language. The Finnish language includes morphophonological variation, in which certain morphemes may appear in different phonological representations depending on the phonological characteristics of the word root (see Comrie 1989). Grammatical knowledge is disciplinary-based—built within the framework of academic

linguistics (see Dielemans and Coppen 2020). While the concept of grammatical understanding has predominantly been applied to language from a descriptive perspective (i.e. describing how the language system works), grammatical understanding could also be applied to prescriptive grammar (i.e. understanding how language norms and grammatical conventions work, for example across different contexts and registers). In the current study, grammatical understanding encompasses both perspectives towards grammar. Grammatical understanding can be implicit (involving language users' naturally developed language intuitions) and explicit (i.e. their verbalizable knowledge of grammar) (Camps and Fontich 2019). It is the explicit type that is at the heart of our study.

A related concept to grammatical understanding is *metalinguistic understanding*, which is a term that has been used in reference to conscious analysis and manipulation of language in a broad sense (e.g. Myhill 2011; Myhill and van Rijt 2023). This implies that while metalinguistic understanding can refer to analyzing and manipulating many aspects of language (e.g. language variation, change, and use), grammatical understanding focuses on the more specific syntactical, morphological, and phonological structures of language. According to Myhill and van Rijt (2023), metalinguistic understanding is explicit by nature, because it is conscious and manifested in stepping back from language use in order to observe and analyze language from a distance. While we have previously acknowledged that grammatical understanding can also manifest implicitly (contrary, it seems, to metalinguistic understanding), the current study exclusively focuses on the explicit dimension of grammatical understanding.

Another related and widely used concept is language awareness, which is defined by the Association for Language Awareness (n.d.) as the 'explicit knowledge about language, and conscious perception and sensitivity in language learning, language teaching and language use'. It is thus a broader concept that can be related to many kinds of understanding about language. While we acknowledge the importance of this broad-ranging understanding, our own study takes grammatical understanding as a more specific focal point.

Reading comprehension

In a now classical study, Kintsch (1998) described reading comprehension as a process in which a reader creates a mental representation of the situation described in the text, also known as a situation model. In a similar vein, Woolley (2011, pp. 15–16) states that reading comprehension is 'a flexible and ongoing cognitive and constructive process' that 'integrates information from the text-based model with information from prior knowledge using inferential processing'. It is a complex set of skills, involving psycholinguistic processing of symbolical representations of letters, morphemes, words, phrases, and sentences. The significance on grammatical understanding within these processes will be discussed in the following section. Reading comprehension also involves connecting these parts of texts to each other and to a wider understanding of themes they discuss and their contexts. This requires linking what is read with reader's background knowledge (Woolley 2011).

Reading skills can be distinguished from reading strategies: while the former can be defined as 'automatic actions that result in comprehension', the latter are 'deliberate, goal-directed attempts to control and modify the reader's effort to decode text, understand words, and construct meanings' (Afflerbach et al. 2008, p. 368). While more traditional views on reading have predominantly focused on written texts, our paper attends to a wider notion

of multiliteracies, according to which reading is defined as interpretation of multimodal texts that are encountered in multiple communication channels and that may combine, for example, letters and visual images (Cope and Kalantzis 2000). Being 'multiliterate' also means taking into account the diverse reality of using language (Cope and Kalantzis 2000).

Relationships between grammatical understanding and reading comprehension

According to a recent meta-analysis, studies have generally observed large correlations between grammatical knowledge and reading comprehension (Zheng et al. 2023). However, studies have differed in the operationalizations of grammatical understanding. In the following, we will discuss grammatical understanding from the viewpoints of morphological and syntactic understanding.

Morphological awareness refers to the processing of morphemes, which are the smallest units of meaning in a language. It has recently been subject to a considerable body of research (Zhang et al. 2023). Many studies have indicated that morphological awareness is connected to reading comprehension (e.g. Bowers et al. 2010, Goodwin and Ahn 2010; Kirby et al. 2012; Zhang et al. 2023). Morphological awareness is also shown to compensate some of the problems in phonological processing, which is considered a factor behind reading difficulties (Casalis et al. 2004; Elbro and Arnbak 1996; Goodwin and Ahn 2010).

According to Perfetti (2007, 2017), the morphological understanding of word structures and formation may contribute to lexical quality, which in turn enhances reading comprehension. Verhoeven and Perfetti (2008) argue that the reading processes differ in more and less 'known' words; the more familiar words are recognized orthographically and directly, whereas rarer and more complex forms need to be divided into their respective morphemes. This might be especially relevant within linguistic contexts such as Finnish, where meanings are largely constructed by adding morphemes into word roots.

Muter et al. (2004) have measured grammatical understanding as a sum variable of syntactic and morphological awareness, concluding that grammatical skills predict reading comprehension in young primary school students. Similar results have been detected in 8- to 9-year-olds (Logan and Language and Reading Research Consortium 2017). Correspondingly, difficulties in reading comprehension have appeared connected to weak grammatical skills (Adlof and Catts 2015; Nation et al. 2004; Tong et al. 2014).

Previous studies have also identified significant correlations between syntactic awareness and reading comprehension (Grabe and Yamashita 2022). Deacon and Kieffer (2018) found that syntactic awareness predicts reading comprehension in 3rd and 4th grade L1 students. A similar effect was found by Brimo et al. (2017) in 9th and 10th grade L1 students. Even young adults have been shown to benefit from discussing the syntactic construction of clauses and sentences as it may improve understanding of complex clauses (Chipere, 2003). Even though L2 learning is beyond the scope of this study, it is worth noting that the connection between syntactic awareness and reading comprehension has been well established within L2 research (Grabe and Yamashita 2022, p. 290–292).

Grabe and Yamashita (2022) point out that grammatical parsing of a text is not a skill that is sometimes used but an inseparable part of reading. This is because (while vocabulary knowledge is surely important in reading) mere chains of words are unable convey discourse meanings without grammatical structures. According to Givon (1995), grammatical cues

provide information on referentiality, temporality, locations, event continuity, modality (e.g. how certain the speaker is) as well as textual coherence in a larger scale. Correspondingly, the reader does not only process words and phrases but also their forms (morphological processing) and their relationships to each other (syntactic processing). Syntactic processing can be demanding also for advanced speakers of a language: some complex structures are not part of even a native speaker's naturally acquired linguistic repertoire, but they are part of a more formal register, which is often introduced to children at schools (Chipere 2003). There also seem to be differences between spontaneous syntactic processing, which is performed implicitly when interpreting a sentence, and a secondary use of syntactic information, which means attending the syntax analytically and deliberately (Funke and Sieger 2012). For example, ninth-grade students have been observed to draw connections between grammatical concepts and the meanings that they create in texts, especially when receiving pedagogical support from a subject-specialist teacher (Marjokorpi 2019, 2022). Even younger students may be assisted in reading children's literature from a grammatical perspective by paying instructional attention to, for example, theme selection or mood and modality (Unsworth 2002). Therefore, we can expect explicit grammatical understanding to be connected to reading comprehension.

The linguistic and educational context of Finland

The current study is set in Finland and pertains to the Finnish language. Unlike most European languages, Finnish is not part of the Indo-European language family; instead, it is a Finno-Uralic language, as are, for example, Estonian and Hungarian. Therefore, the language structures of Finnish differ from the Indo-European languages in some respects. Firstly, Finnish is a rather synthetic language which tends to use inflectional suffixes to mark grammatical relations. Therefore, morphology can be said to have a bigger role in understanding the Finnish language than in understanding more analytic languages, like English or Dutch. Secondly, in connection with the synthetic nature of Finnish, the language has a relatively free word order (Vilkuna 1989). Thirdly, there are some phonological types of grammar rules, also addressed in the assessment data analyzed in this paper. For example, the Finnish language distinguishes front vowels (y, ä, ö) from back vowels (u, a, o), and the two vowel types cannot appear together in the same word (root), except in some loan words and compounds. This phenomenon is called vowel harmony (see Gordon 2006), and it affects, for example, the phonological characteristics of case endings.

School grammar teaching in Finland has traditionally focused on identifying and naming different inflectional forms, parts of speech, and clause constituents in a rather decontextualized way. This pedagogy has already been heavily critiqued for decades, and there have been strong aspirations towards more functional and meaningful grammar pedagogies, at least on the level of curriculum design and subject didactics (Marjokorpi 2023b). Functional teaching of knowledge about language is also emphasized in the National Core Curriculum for basic education (NCC 2016), and it does not require teaching any specific grammatical (meta)concepts when discussing language structures. The situation has been different in many Anglophone countries where the curricula temporarily abandoned grammar in the 1960s and have more recently started to reintroduce it (e.g. Myhill and Watson 2014). Finnish textbooks, on the other hand, seem to have continued with a rather traditional

terminology (Marjokorpi 2023c) – in this sense, the Finnish context seems to be little different from most international contexts, where traditional terminology and teaching approaches are also commonplace (Van Rijt et al. 2019). While there is evidence that grammar is less emphasized now than, for example, the mid-20th century (e.g. Marjokorpi 2023c), it seems that most Finnish students are still taught the basics of the Latin-based system of classifying words, inflections, and sentence constituents. It is uncertain whether the emphasis on grammar has been steadily declining or if there have been ups and downs. This uncertainty is arguably related to the high level of teacher autonomy and the lack of standardized tests in Finnish primary and lower secondary education. National assessment studies (e.g. Lappalainen 2004; Harjunen and Rautopuro 2015; Kauppinen and Marjanen 2020) have observed no specific changes in students' grammatical skills and knowledge across time. In this sense, the Finnish situation seems to be somewhat different from other educational contexts, in which declines in grammatical knowledge have been observed (e.g. Alderson and Hudson 2013).

When it comes to literacy skills, there is much more research – and much more worry. While the first international PISA and PIRLS studies celebrated Finnish students as one of the most literate in the world, recent years have witnessed a decline in results. The most recent reports (Leino et al. 2023; Leino et al. 2019) have concluded that the differences in literacy skills have increased, the number of high-achieving readers has declined, and the number of low-achieving readers has risen significantly. The direction has been a topic of public debate, and much research is devoted in finding the reasons for the decline. Studies have observed that reading skills are affected by the socio-economic status of students' families, their reading habits, the expansion of digital media environment, and students' attitudes (e.g. self-efficacy and enjoyment) towards reading (Leino et al. 2023). Our paper participates in the discussion by examining the effect of explicit grammatical understanding alongside some of these factors on students' reading skills.

Data and methods

National assessment

Our data come from a national assessment of L1 and literature learning results, conducted by the Finnish Education Evaluation Centre (FINEEC). A total of 6,044 ninth-grade students (ca. 15-year-olds; 2,798 girls, 2,715 boys, 48 of other gender, 483 with information missing) participated in the study in 2019. According to Kauppinen and Marjanen (2020), who conducted the original assessment study, the sampling was randomized on the school level, and in big schools, also on the individual level. Different regions and municipality types in Finland were well represented in the sample.

One of the authors of this study has used the same data set before to examine the relationship between grammatical understanding and writing skills (Marjokorpi 2023a). In the current study, we focus on grammatical understanding and reading. The FINEEC has granted the researchers an official permission to use the data and given preliminary guidance in its use (see also Marjokorpi 2023a). In all other regards, this study has been conducted independently and with no affiliation or exchange of ideas with FINEEC.

The students completed the test in two parts during school lessons in an online environment. The first part included writing items and a background survey, which dealt with, e.g. attitudes towards studying L1 and literature, home background, and studying and

reading habits. The second part assessed multimodal text interpretations skills and knowledge about language, including grammatical understanding. Due to the confidentiality contract and the plans of reusing some of the test items, the exact items cannot be reprinted in this paper. However, in the following, we will describe the measurement of reading and grammatical understanding, as well as other relevant variables.

Reading comprehension in the assessment

In alignment with the Finnish L1 and literature curriculum, the FINEEC assessment builds on a multimodal understanding of reading literacy. Therefore, the original assessment included six types of texts to be analyzed and interpreted by students: (1) an online forum discussion, (2) a radio discussion (in an audio format), (3) book covers, (4) a campaign poster, (5) a short story extract, and (6) a poem. The assessment also had 8 items concerning general knowledge about literary genres (7). We decided to exclude the listening items (2) for reasons of validity (i.e. listening constitutes a different activity than reading). We also excluded the general knowledge items (7) which did not measure reading literacy. All other items were included. They consisted of texts that could be read on a screen, including pictures and photos, reflecting how international assessments (e.g. PISA) measure reading comprehension. The students answered both open-ended and multiple-choice questions about, for example, the communicative functions and ways of meaning-making in the campaign poster or the interpretations of metaphoric language in book covers (see also Kauppinen and Marjanen 2020, p. 49–50). The open-ended questions were assessed by teachers according to pre-established criteria, and a randomly selected 10% of the answers were re-assessed by FINEEC professionals. A statistical analysis of these two assessment results confirmed that the reliability of the assignments was good or satisfactory (Kauppinen and Marjanen 2020, p. 37).

Grammatical understanding in the assessment

The operationalization of grammatical understanding reflects our definition of the concept (see above). The sum variable of grammatical understanding consists of 14 items, 11 of which were multiple choice and 3 were open-ended items. 13 items dealt with syntactical, phonological, and morphological language forms and their functions, and they were each worth one point. One open-ended item measured understanding of written standard language norms, and it was worth 2 points. The open-ended items were scored by teachers according to pre-established guidelines. The multiple-choice items were scored automatically.

The background variables in the assessment

We also took into account various relevant background variables that can be theoretically assumed to predict reading comprehension, and therefore had to be controlled for. Previous studies (e.g. Leino et al. 2019, 2021, 2023) have indicated that the socio-economic background of the family is a strong predictor of a student's literacy skills. In our data, the socio-economic background was measured by asking the students to report the highest level of schooling of both parents. In addition, reading habits have been shown to predict

reading skills (Leino et al. 2023). In our data, reading habits were measured as the number of books read per month. Attitude variables included liking the L1 school subject, self-efficacy, and considering studying L1 useful. They were compound from multiple Likert-scale questions. We also took into account the students' writing scores which were measured using two writing assignments: the students first wrote an argumentative text (max. 14 points) about one of preassigned topics and then a reflective text (max. 12 points) in which they discussed, e.g. their writing processes and themselves as writers (see also Kauppinen and Marjanen 2020; Marjokorpi 2023a).

Methods

Statistical analyses

Before we examined whether grammatical understanding could predict reading comprehension, we checked descriptive statistics and examined correlations between the relevant variables described in more detail above (e.g. parents' education, homework frequency, liking the L1 subject). For certain predictors, which were composed of various subscales, we assessed internal consistency and reliability by computing Cronbach's alpha.

We then carried out a linear regression analysis with reading comprehension as the outcome variable and the other significantly correlating variables as predictors. Since the students from the assessment were nested in different schools, we determined the intraclass correlation (ICC) to assess the variance that could be explained by students belonging to different schools, following the approach proposed by Sommet and Morselli (2017). The ICC was 0.092, indicating that schools could account for 9.2% of the variance in reading comprehension scores. Given that even small ICC's are sufficient reason to use a mixed model (Nezlek 2008), we employed a multilevel approach in the regression analyses, following a step-up model building strategy (Sommet and Morselli 2017). In the regression model, we assumed a random effect of school on intercept (i.e. some schools might have stronger students than others) and a fixed effect of the other variables (i.e. the predictors are assumed to have an equal effect for all students). A Likelihood Ratio Test (LRT) for random effects revealed a statistically significant improvement in model fit for a model that includes school, $\chi^2(1)=130$, $p<.001$, compared to a model that does not include school as a random factor.

We started with a basic intercept-only model (Model 1), in which the intercept (mean score) of reading comprehension was estimated. We then added variables related to students' background (Model 2), namely their mother's education, their father's education, frequency of doing homework, and book reading. These variables were assumed to contribute to reading comprehension and were seen as necessary controls before grammatical understanding could be taken into account. We subsequently added predictors related to the L1 subject (Model 3): students' liking of the L1 subject, the extent in which they felt the L1 subject was useful and their self-efficacy in the L1 subject. Given the strong relationships between reading and writing (see e.g. Schoonen 2019), we added students' writing scores in the next model (Model 4). Finally, we included grammatical understanding in the model (Model 5), to see if grammatical understanding could still be predictive of reading comprehension once we controlled for all other relevant variables. For the multilevel modeling, we used Jamovi's (2022) GAMLj package (Gallucci 2019).

Results

Descriptives

Table 1 provides some descriptive statistics for the variables taken into account in the study. The Table reveals that for each variable, some data were missing as a result of students leaving questions blank. Figure 1 shows the distributions of these data, revealing that some of these variables are not normally distributed; some variables like Reading comprehension or Usefulness of L1 are skewed. This does not pose a significant problem in multilevel regression, however, if the distribution of residuals is roughly normal, as is the case for our dataset (cf. Shaw and Flake 2023, Chapter 12).

The variables of Book Reading and Homework show opposite patterns: the self-reported number of books read per month is low for most students, whereas many students do claim that they faithfully do their homework. These distributions are in line with what one might expect when questioning students about these matters, but they also show that the measures are self-reports (a point to which we will return in the discussion section).

As previously outlined, a few variables were comprised of separate items. To check that they measured one construct, we calculated Cronbach's alpha for these variables. This applies to the following variables: grammatical understanding (Cronbach's $\alpha = 0.74$), Liking L1 (Cronbach's $\alpha = 0.85$), usefulness of L1 (Cronbach's $\alpha = 0.86$) and self-efficacy in the L1 subject (Cronbach's $\alpha = 0.80$). As all of these values are above 0.70, we have assumed that these variables are in fact measuring one underlying construct (cf. DeVellis 2003). See Appendix A for the statements related to Liking L1, Usefulness of L1, Self-efficacy, Book reading, and Homework.

The participants were asked about their parents' educational level, which has in previous studies appeared as a significant predictor of reading literacy (e.g. Leino et al. 2023). Almost one fourth of the students reported that their mother had a university degree, and nearly as many reported of a vocational degree. Mother's education was reported unknown by 13% of the participants. According to the student responses, their fathers tended to be slightly less educated compared to the mothers: 15% of the fathers were reported to have a university education, and 32% of them to have a vocational degree. 15% of the students did not know about their father's education

In the next step, we created a Pearson's correlation matrix to explore correlations between these variables (Figure 2). Any variables that correlated significantly to reading comprehension were added to the regression models. Figure 2 shows that all of the variables correlate to all of the other variables significantly, so they were all taken into account in the regression analyses. The highest correlations were found between reading comprehension and writing (0.65), between reading and grammatical understanding (0.63), and between writing and grammatical understanding (0.57).

Table 1. Descriptive statistics for the relevant variables.

	N	Missing	Mean	Median	SD	Minimum	Maximum
Reading comprehension	5430	614	38.38	40.00	9.36	3	56
Grammatical understanding	5454	590	8.64	9.00	3.12	0	15
Book reading	5602	442	1.61	1.00	0.91	1	5
Homework	5618	426	4.11	5.00	1.11	1	5
Liking L1	5620	424	2.95	3.00	1.00	1	5
Usefulness of L1	5625	419	3.89	4.00	0.89	1	5
Self efficacy	5626	418	3.15	3.20	0.65	1	5
Writing	5564	480	14.00	14.00	5.99	0	26

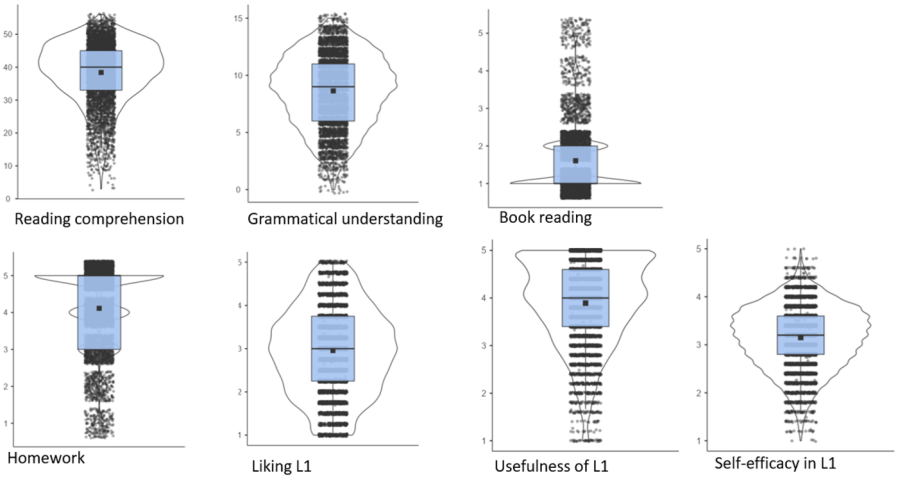


Figure 1. Violin plots showing the distribution of each of the study's variables. Squares represent mean scores; each dot represents an individual observation.

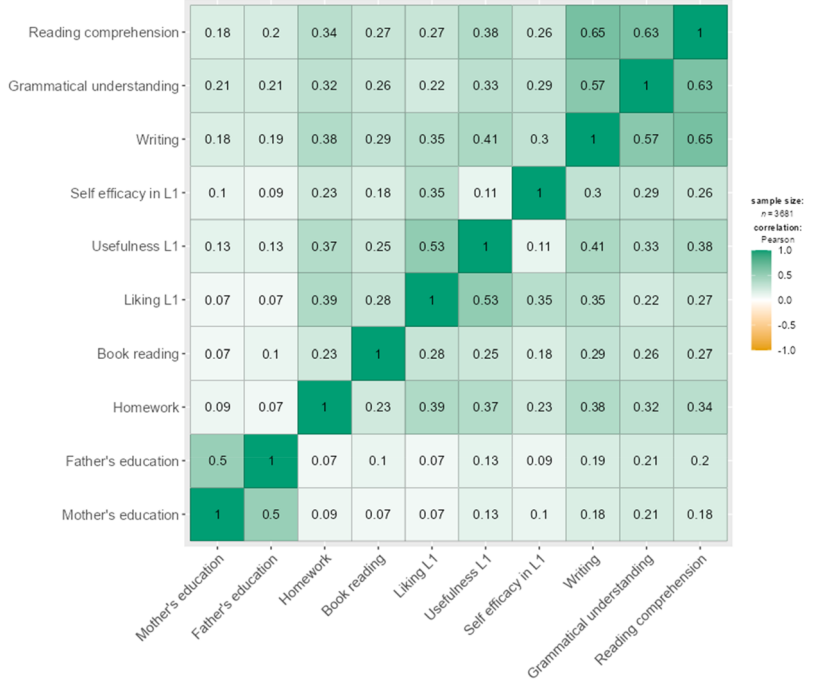


Figure 2. Pearson's correlations between the different variables. All correlations are positive and significant, with darker shades of green reflecting higher levels of positive correlations.
Note: All reported correlations are significant at the $p < .001$ level.

Multilevel regression analyses

Following a stepwise build-up strategy for the multiple regression analyses (Restricted Maximum Likelihood), we started from a first model (M1) and added more predictors in four subsequent steps. Table 2 shows the model information for each model, illustrating that Model 5 is the best model to describe these data. Each model is better than the last, showing lower AIC and BIC values, lower-2LL values and larger R^2 values (reflecting better models). Given this state of affairs, we will only report on the specifics of Model 5. Model 5 explains 52% of the variance in the dataset. This means that when controlling for relevant background variables, grammatical understanding still has predictive power on top of those. Model 5, containing grammatical understanding on top of all other variables, explains 7% more of the variance than Model 4. The syntax of this model is as follows: Reading comprehension ~ 1 + Mother's education + Father's education + Homework + Book reading + Liking L1 + Usefulness L1 + Self efficacy + Writing + Grammatical understanding + (1 | SchoolID).

Table 3 contains the specific information belonging to model 5, which is based on 3,681 students' data (based on listwise deletion), spread over 114 different schools. The model shows that all variables, except for Mother's education, are significant predictors of reading comprehension. The model estimates reading comprehension scores and predicts how much reading comprehension improves (or declines) as the other variables increase. For instance, for every additional point a student scores on grammatical understanding, reading comprehension scores increase by an estimated 1.02 points if all other predictors remain equal (see 'Estimate'). Similarly, for every additional point scored for writing, reading comprehension scores go up by 0.57. The β column from Table 3 reports standardized regression coefficients (where the variables have been converted to Z scores), making it possible to compare the weight of the different predictors in relation to one another (i.e. to assess the

Table 2. Model comparisons showing how each model improves compared to the last.

	AIC	BIC	−2LL	Δ − 2LL	R^2
Model 1	39375.586	39395.968	19685.084	–	–
Model 2	26248.175	26302.004	13122.202	6562.88	0.18
Model 3	25848.956	25926.813	12922.271	199.931	0.25
Model 4	24418.360	24510.442	12210.040	712.231	0.45
Model 5	23779.239	23883.291	11892.380	317.660	0.52

Table 3. Fixed effects parameters for model 5 (95% CI).

				95% Confidence interval				
Names	Estimate	β	SE	Lower	Upper	df	t	p
(Intercept)	38.8979	−0.02094	0.1923	38.5210	39.2748	105	202.287	< .001
Mother's education	0.0445	0.00621	0.0930	−0.1378	0.2268	3661	0.478	0.632
Father's education	0.1954	0.02773	0.0924	0.0143	0.3765	3671	2.115	0.034
Homework	0.3693	0.04233	0.1136	0.1467	0.5919	3668	3.252	0.001
Book reading	0.4141	0.04228	0.1180	0.1828	0.6454	3657	3.509	< .001
Liking L1	−0.3498	−0.03853	0.1332	−0.6108	−0.0888	3671	−2.627	0.009
Usefulness of L1	0.9497	0.09096	0.1497	0.6563	1.2431	3662	6.344	< .001
Self efficacy in L1	0.4953	0.03560	0.1773	0.1479	0.8427	3671	2.794	0.005
Writing	0.5744	0.37210	0.0233	0.5287	0.6202	3648	24.626	< .001
Grammatical understanding	1.0173	0.34477	0.0424	0.9343	1.1003	3671	24.018	< .001

Note: SE shows the standard error of the estimate, i.e. the standard deviation of the residuals

relative importance of the predictors). The standardized regression coefficient for Writing is 0.37, indicating that, for every one standard deviation increase in Writing, the dependent variable (reading comprehension) is expected to increase by 0.37 standard deviations. Similarly, the standardized coefficient for grammatical understanding is 0.34, suggesting that, for every one standard deviation increase in grammatical understanding, reading comprehension is expected to increase by 0.34 standard deviations. The difference in coefficients suggests that Writing has a slightly stronger impact on the dependent variable compared to Grammatical Understanding, although they are both strong predictors.

All significant variables seem to positively contribute to reading comprehension, except for Liking L1, which seems to have a negative influence on reading comprehension as indicated by the negative estimate (−0.35). VIF diagnostics indicated no substantial multicollinearity issues, as all VIF values were well below 10, and even all below 2. This indicates that the predictors in the model are sufficiently independent.

Appendix B shows the effect of different schools on the relationship between grammatical understanding and reading comprehension.

6. Discussion

The purpose of our study was to examine the relationship between grammatical understanding and reading literacy skills in Finnish 9th grade students. Using large-scale national assessment data and multilevel quantitative analyses, we found out that explicit grammatical understanding explained a large proportion of the variance in reading comprehension. The effect remained even when the effects of school, family SES, homework and book reading habits, attitudes towards studying L1, and writing ability were controlled for. Understanding how language works and being able to make explicit metalinguistic observations thus seem to be closely connected to reading comprehension. Our data (Table 3) shows that the connection between this type of metalinguistic understanding and reading is almost as tight as the connection between writing ability and reading comprehension (cf. Schoonen 2019). The observations are in concert with previous studies that highlight the inseparability of morphological and syntactic processing from reading (see e.g. Grabe and Yamashita 2022).

These observations encourage researching the benefits of explicit grammar teaching on reading comprehension more closely, preferably using different kinds of study designs. As we observed students' grammatical understanding in a summative manner, one further strand of research could be to discuss how grammar *instruction* is and can be linked to reading. Even though some qualitative studies have observed how students make linguistic observations when reading (media) texts (Marjokorpi 2019, 2022; Watson and Newman 2017), more research is needed about how this metalinguistic reasoning can be taught and fostered.

Our study, as well as many previous studies, has also been conducted in a very specific educational and linguistic context, making it difficult to generalize the findings to other contexts. Even though a recent meta-analysis has suggested that language type may not be a significant factor in the correlation between grammatical knowledge and reading comprehension (Zheng et al. 2023), more studies across different countries are needed to gain a deeper understanding of the relationships between reading comprehension and grammatical understanding. For other languages that rely less on combining morphemes to mark syntactic relations than Finnish, the relationship between reading comprehension and

explicit grammatical understanding may be different. After all, relationships between morphological awareness and reading comprehension are well established (Bowers et al. 2010; Zhang et al. 2023). Part of the result we found may stem from the fact that Finnish is comparably agglutinative. It would therefore be recommendable to replicate the current study in languages that are less agglutinative and more analytic, so we can establish whether similar results can be obtained for other parts of grammar, such as syntax.

Because we were prohibited from reprinting the assessment tasks, other researchers will be unable to directly replicate our study. We would therefore recommend that future studies attempt to work with tasks and materials that do not have this limitation.

Another limitation relates to the operationalization of some of the core concepts of this study. Even though we defined grammatical understanding and explained how it has been operationalized and assessed in the FINEEC data collection, the term 'grammatical understanding' or equivalents thereof (e.g. grammatical knowledge, grammatical awareness, metalinguistic understanding) are used differently in the field of educational linguistics (Myhill and van Rijt 2023). Therefore, future studies need to pay attention to this aspect. In the present study, we have perceived grammatical understanding as a multi-faceted variable, including morphological, syntactic, and phonological components, and both encompassing descriptive and (to a much lesser extent) prescriptive knowledge. This means we cannot make claims about what areas of grammar support reading comprehension. In addition, the concept of reading comprehension can be understood differently (Woolley 2011). Our interpretation stems from the current Finnish curriculum, in which reading comprehension is perceived as the ability to interpret multimodal texts (NCC 2016), acknowledging the importance of multimodality in reading comprehension (Unsworth 2014).

It is also good to note that students might not always have reliable perceptions of their parents' schooling (e.g. Avvisati 2020), and self-reports about doing homework or reading books can be affected by social desirability bias (e.g. Randall and Fernandes 1991). Therefore, our study might have an overestimated view on students' studying and reading habits. At the same time, it shows that even with this potentially optimistic picture, a clear effect of grammatical understanding can be found. This implies that a decrease in the scores of these other variables could lead to a more pronounced effect of grammatical understanding.

When interpreting summative cross-sectional studies that measure learning results, a crucially important question is how the students have been taught. In the theoretical chapter of the present paper, we discussed grammar teaching in Finland (see also Nupponen et al. 2019). It might be interesting to see if differently taught grammar relates to literacy skills in a similar way as we observed in this study (cf. Batalha 2019).

While we observed a clear connection between grammatical understanding and reading comprehension, the analyses do not provide information about the directionality of the relationship. This is why the results should not be interpreted causally. Based on previous research, there are good reasons to assume that the two sets of knowledge mutually affect each other, and are also affected by other variables, such as reading habits and attitudes. In fact, the relationship between reading comprehension and grammatical understanding has also been examined the other way around, where more reading leads to improved grammatical mastery, at least in the domain of prescriptive grammar (Favier and Huettig 2021). Even though our study could not make any causal claims pertaining to the relationship between reading comprehension and grammatical understanding, it did show that explicitly understanding how grammar works also plays a crucial role in reading comprehension.

This may mean that certain forms of explicit grammar education can – potentially – have a positive side-effect on reading comprehension, although more research is needed before such claims can be made. Nevertheless, even if future research would show links between explicit grammar teaching and reading comprehension, we would not recommend replacing effective reading comprehension practices (Duke et al. 2010) with explicit grammar instruction. Rather, it may be that certain forms of explicit grammar teaching can contribute to reading comprehension indirectly, on top of instructional practices that have proven beneficial for fostering reading comprehension. Future research should examine this further.

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Appendix A. Descriptions of study variables

This appendix contains information about the background variables, originally collected by FINEEC in 2019. The statements used in composing the variables have been translated from Finnish by the first author. For the Finnish originals, see Kauppinen and Marjanen (2020, p. 93). Inversed variables have been marked with an asterisk (*).

Liking L1 refers to how much the students like studying Finnish language (L1) and literature. The variable has been constructed from the Likert-scale responses to the following statements:

- I like the lessons of L1 and literature.
- L1 and literature is one of my favorite subjects.
- I am happy to study L1 and literature.
- Usually, we have interesting exercises in L1 and literature.
- * L1 and literature is a boring subject.

The mean score of the Liking L1 in all students was 2.96 (Kauppinen and Marjanen 2020, p. 92).

Self-efficacy in L1 refers to how much confidence the students have in their own capabilities in studying Finnish language (L1) and literature. The variable has been constructed from the Likert-scale responses to the following statements:

- L1 and literature is an easy subject.
- *It is impossible for me to gain good results in L1 and literature.
- I think I am good at L1 and literature.
- *Many things in L1 and literature are difficult.
- I can manage even difficult exercises in L1 and literature.

The mean score of Self-efficacy in L1 in all students was 3.42 (Kauppinen and Marjanen 2020, p. 92).

Usefulness of L1 refers to how useful the students find studying Finnish language (L1) and literature. The variable has been constructed from the Likert-scale responses to the following statements:

- In my future studies, I will need the skills and knowledge of L1 and literature.
- The skills and knowledge of L1 and literature are needed in situations of everyday life.
- *In the future, I do not need much of what we have studied in L1 and literature so far.
- I believe I will need the skills and knowledge of L1 and literature in the working life.
- I think it is important to know L1 and literature.

The mean score of Usefulness of L1 in all students was 3.90 (Kauppinen and Marjanen 2020, p. 92).

Book reading refers to how many books the students estimate they usually read per month. The response options of this multiple-choice item were:

- No books
- 1 Book
- 2 Books
- 3 Books
- 4 Books or more

More than half (59%) of the students responded reading zero books per month; 29% reported reading one, 7% reading two, 2% reading three, and 3% reading four or more (Kauppinen and Marjanen 2020, 122).

Homework refers to how regularly the students do their homework. The response options of this multiple-choice item were:

- Never
- Only seldom

- Sometimes
- Often
- Nearly always

Three out of four students ($N = 4{,}160$) reported doing their homework often or nearly always (Kauppinen and Marjanen 2020, 96).

Mother's and father's education refer to parents' educational levels according to student's knowledge. The educational level of both parents was asked separately, using similar multiple-choice questions. The response options were:

- Basic education
- Vocational education
- Secondary education
- University of applied sciences
- University
- None of above
- I do not know.

The distribution of the student responses is reported in our paper.

Appendix B. Visualization of the effect of school on the relationship between grammatical understanding and reading comprehension

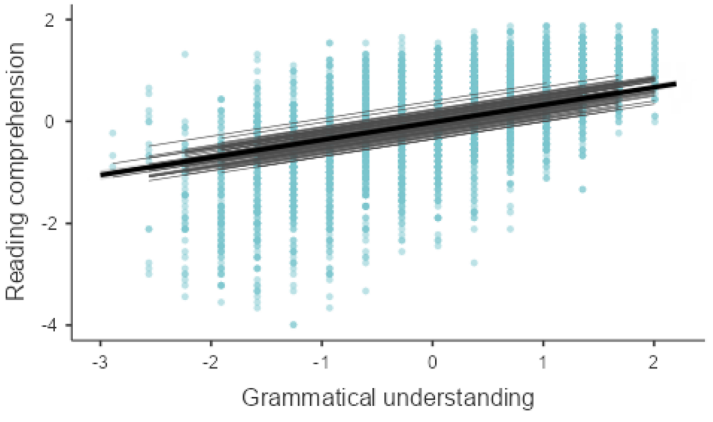


Figure B1. Visual representation of the effect of school on the relationship between reading comprehension and grammatical understanding. Each line represents one of the 114 schools; each dot represents an observation.